Supplementary: subdistribution HR (Fine–Gray) by progression pattern (reference: IH-NL)

Fine–Gray via survival::finegray with stratum-specific censoring weights; robust SE

	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.08 (0.88–1.34)	1.50 (0.78–2.89)	1.16 (0.75–1.81)	0.98 (0.51–1.89)
IH-NL	Reference	Reference	Reference	Reference
IH-TG+NL	1.49 (1.08–2.07)*	1.81 (0.71–4.66)	1.12 (0.59–2.14)	0.80 (0.27–2.37)
EH-TG	1.09 (0.75–1.59)	3.32 (1.59–6.96)**	1.39 (0.78–2.47)	0.83 (0.26–2.66)
EH-NL	0.71 (0.54–0.93)*	2.79 (1.51–5.15)**	1.91 (1.25–2.92)**	1.63 (0.87–3.02)
EH-TG+NL	0.82 (0.42–1.59)	2.59 (0.74–9.09)	1.94 (0.84–4.49)	3.20 (0.89–11.54)
Sys-TG	1.03 (0.60–1.75)	1.41 (0.38–5.24)	3.06 (1.60–5.83)***	NE (1 event)
Sys-NL	0.74 (0.49–1.10)	4.36 (2.02–9.42)***	2.69 (1.58–4.55)***	NE (2 events)
Sys-TG+NL	0.99 (0.74–1.32)	2.08 (1.03–4.20)*	2.74 (1.74–4.31)***	0.78 (0.30–2.00)
Overall Wald P (df=8)	**P = 0.004**	**P = 0.003**	**P < 0.001**	**P = 0.265**

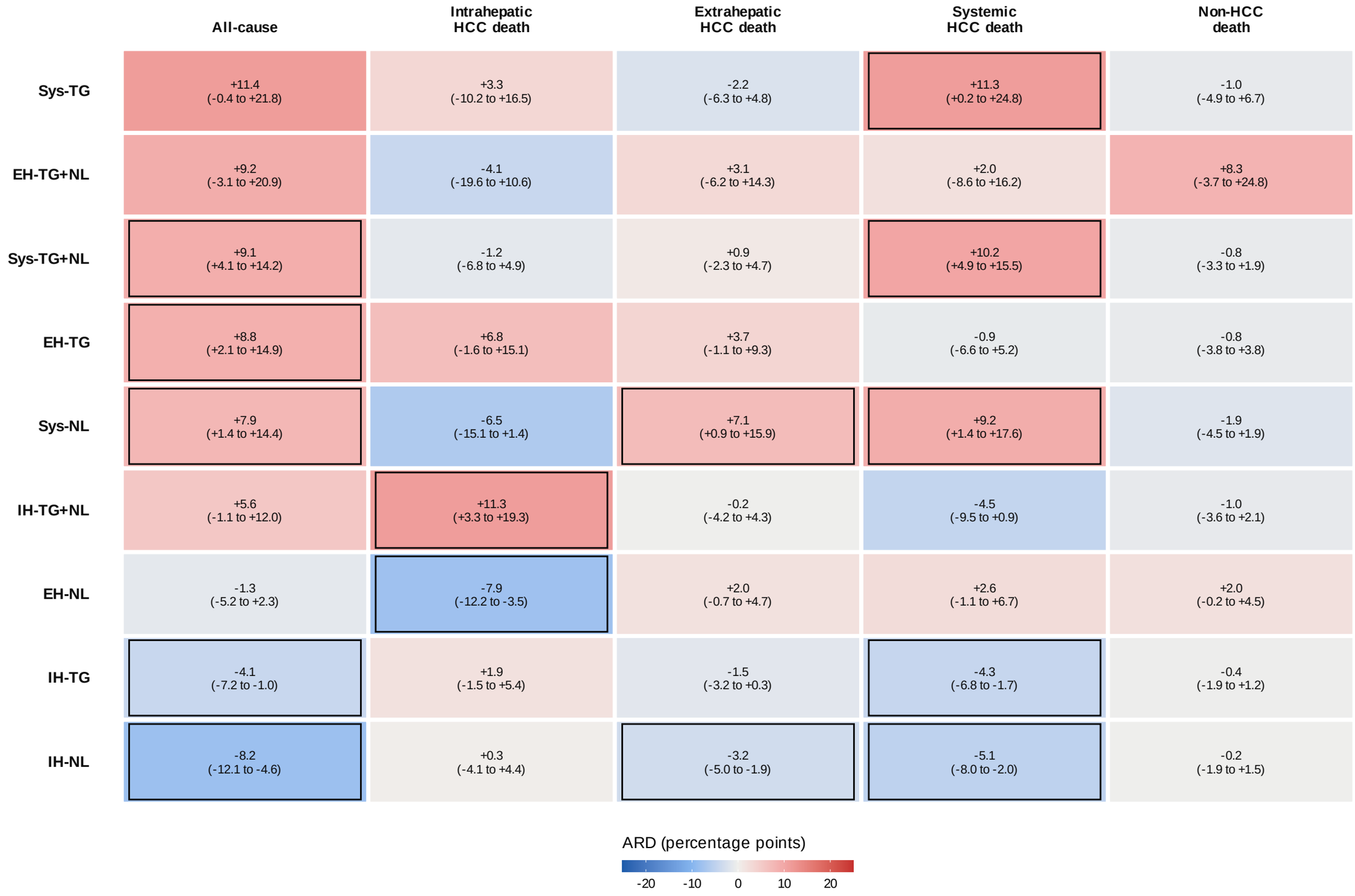
$\log_2(\text{sHR})$

-2 -1 0 1 2

Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: absolute risk difference at 12 months vs population-standardised average

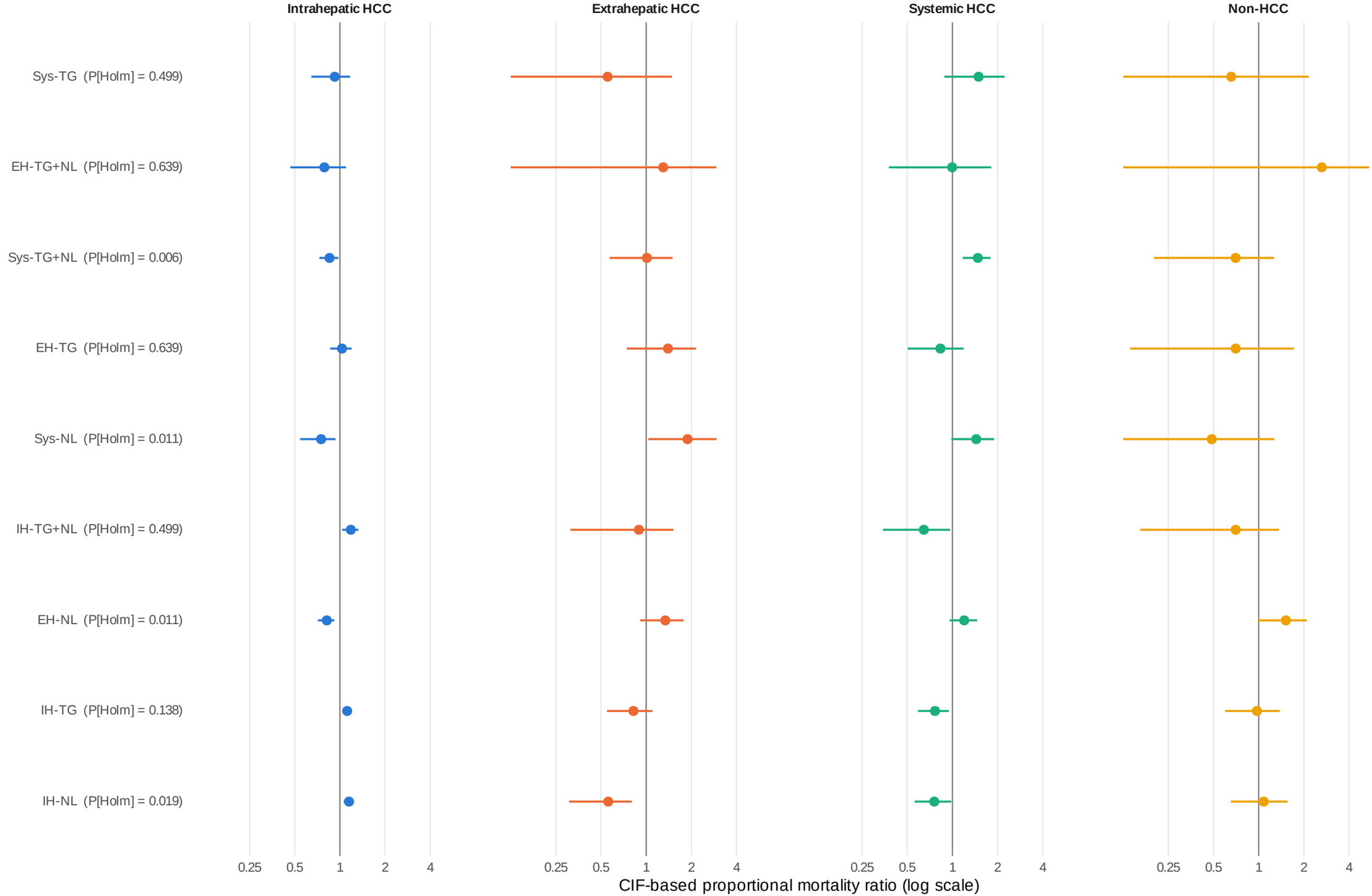
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: cause-of-death enrichment (PMR) at 12 months

PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).

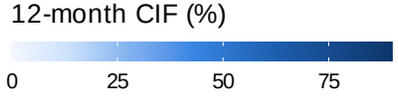


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

Supplementary: unadjusted 12-month cumulative incidence (Aalen–Johansen)

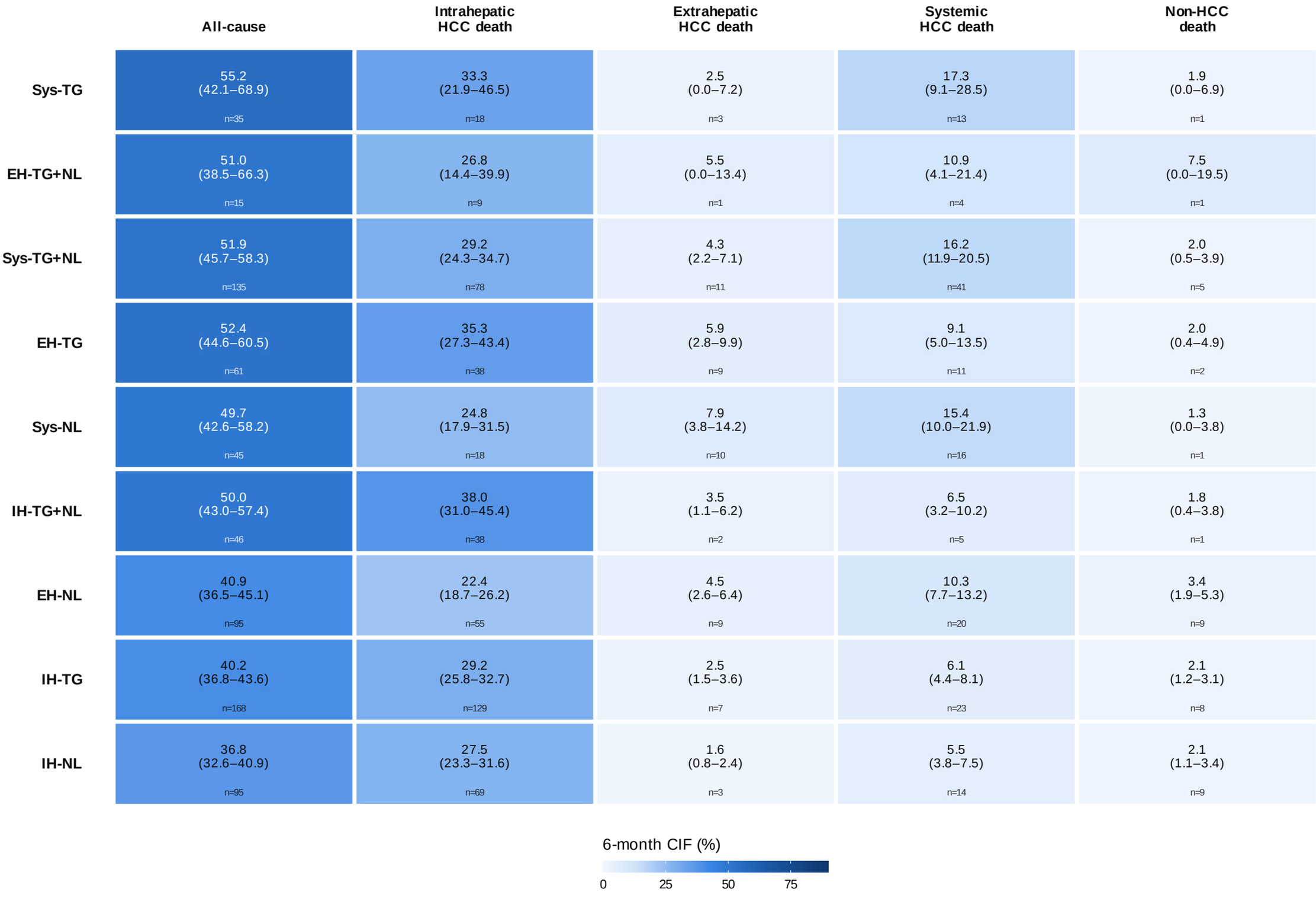
% (95% CI); all-cause = 1 − Kaplan–Meier; n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	78.7 (62.3–88.0) n=36	39.0 (27.1–56.1) n=18	6.3 (2.1–18.9) n=3	31.1 (20.1–48.1) n=14	2.4 (0.3–16.3) n=1
EH-TG+NL	74.0 (51.2–86.2) n=21	35.0 (21.2–57.8) n=10	7.2 (1.9–27.3) n=2	20.9 (10.3–42.8) n=6	10.9 (3.7–31.7) n=3
Sys-TG+NL	79.5 (72.7–84.6) n=154	45.4 (38.9–53.1) n=88	6.8 (4.0–11.4) n=13	24.2 (18.8–31.1) n=47	3.1 (1.4–6.9) n=6
EH-TG	63.4 (53.1–71.4) n=72	36.4 (28.6–46.4) n=42	10.0 (5.7–17.5) n=11	13.5 (8.4–21.6) n=15	2.7 (0.9–8.1) n=3
Sys-NL	81.0 (69.2–88.2) n=63	33.0 (24.1–45.4) n=26	16.8 (10.2–27.7) n=13	28.5 (19.9–40.6) n=22	2.7 (0.7–10.6) n=2
IH-TG+NL	73.0 (62.3–80.7) n=74	53.8 (44.7–64.6) n=55	6.2 (2.9–13.6) n=6	9.9 (5.5–17.8) n=10	3.2 (1.0–9.7) n=3
EH-NL	62.9 (56.2–68.6) n=159	29.3 (24.2–35.5) n=76	7.3 (4.7–11.4) n=18	19.7 (15.3–25.5) n=48	5.9 (3.6–9.6) n=15
IH-TG	66.8 (61.7–71.3) n=269	47.4 (42.7–52.6) n=193	5.0 (3.2–7.7) n=19	9.6 (7.1–13.1) n=38	4.0 (2.5–6.5) n=16
IH-NL	49.8 (43.9–55.1) n=161	33.4 (28.6–39.0) n=109	2.6 (1.3–5.1) n=8	9.2 (6.5–13.0) n=29	4.7 (2.8–7.7) n=15



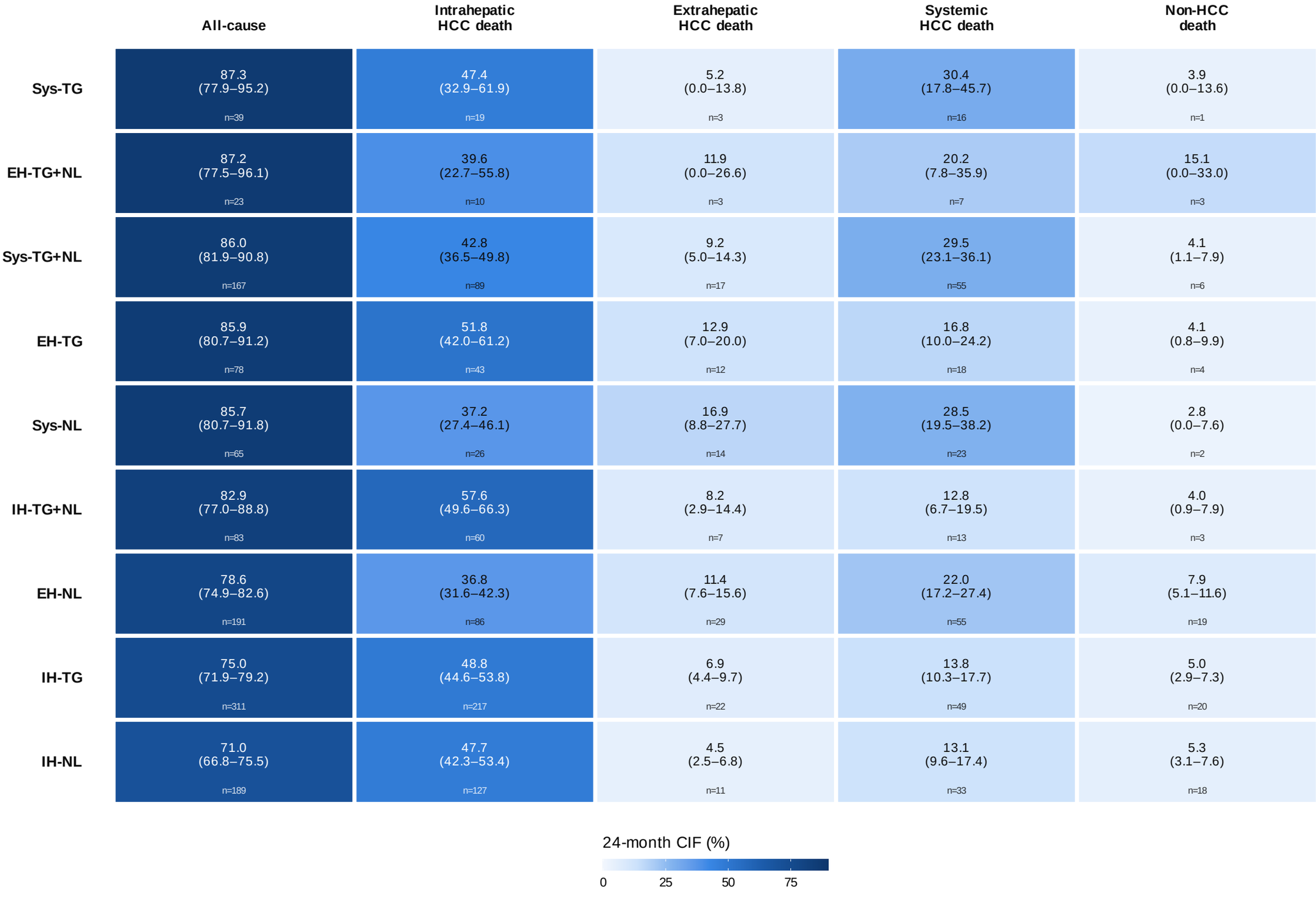
Supplementary: adjusted 6-month CIF after first progression

Same models as Figure 4.4A; n = events within 6 months



Supplementary: adjusted 24-month CIF after first progression

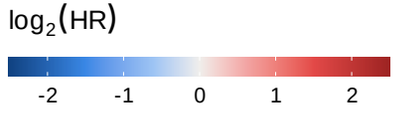
Same models as Figure 4.4A; n = events within 24 months



Supplementary: Firth-penalised HR / csHR (reference: IH-NL)

coxphf (Firth-penalised partial likelihood, Wald CI from penalised information); treatment group entered as covariate (coxphf has no strata)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.17 (0.98–1.40)	1.10 (0.89–1.37)	1.69 (0.87–3.26)	1.18 (0.75–1.84)	1.01 (0.52–1.95)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.55 (1.20–2.00)***	1.59 (1.17–2.16)**	2.58 (1.05–6.34)*	1.46 (0.76–2.77)	1.07 (0.37–3.13)
EH-TG	1.81 (1.39–2.35)***	1.55 (1.09–2.20)*	4.47 (2.14–9.34)***	2.06 (1.16–3.66)*	1.19 (0.40–3.55)
EH-NL	1.26 (1.04–1.54)*	0.84 (0.64–1.11)	3.05 (1.63–5.72)***	1.96 (1.27–3.01)**	1.64 (0.85–3.16)
EH-TG+NL	1.77 (1.14–2.76)*	1.24 (0.65–2.35)	4.85 (1.42–16.60)*	2.37 (1.03–5.45)*	5.07 (1.48–17.30)**
Sys-TG	2.06 (1.45–2.92)***	1.62 (0.99–2.63)	2.23 (0.65–7.60)	4.05 (2.19–7.48)***	NE (1 event)
Sys-NL	1.75 (1.32–2.31)***	1.08 (0.71–1.62)	6.18 (2.90–13.19)***	3.34 (1.95–5.74)***	NE (2 events)
Sys-TG+NL	1.80 (1.46–2.22)***	1.30 (0.99–1.71)	3.13 (1.53–6.42)**	3.57 (2.30–5.53)***	1.17 (0.45–3.03)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.002**	**P < 0.001**	**P < 0.001**	**P = 0.303**

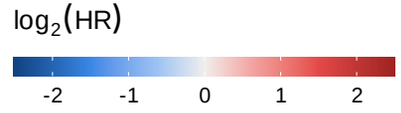


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR excluding undetermined deaths

n = 1778 (8 deaths with unclassified/unknown/missing cause removed)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.16 (0.97–1.39)	1.10 (0.89–1.37)	1.68 (0.87–3.26)	1.17 (0.75–1.84)	1.02 (0.53–1.95)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.58 (1.22–2.04)***	1.64 (1.21–2.23)**	2.49 (1.00–6.22)	1.39 (0.73–2.67)	0.99 (0.33–2.97)
EH-TG	1.80 (1.38–2.34)***	1.56 (1.09–2.22)*	4.41 (2.11–9.23)***	2.00 (1.11–3.58)*	1.07 (0.35–3.29)
EH-NL	1.24 (1.02–1.52)*	0.83 (0.63–1.09)	2.97 (1.58–5.58)***	1.95 (1.26–3.00)**	1.66 (0.87–3.18)
EH-TG+NL	1.70 (1.09–2.66)*	1.12 (0.58–2.16)	4.04 (1.11–14.75)*	2.29 (0.98–5.39)	4.05 (1.13–14.54)*
Sys-TG	2.01 (1.41–2.86)***	1.53 (0.94–2.50)	1.88 (0.52–6.74)	4.07 (2.20–7.54)***	NE (1 event)
Sys-NL	1.70 (1.28–2.25)***	1.02 (0.67–1.54)	5.79 (2.69–12.43)***	3.28 (1.91–5.64)***	NE (2 events)
Sys-TG+NL	1.77 (1.44–2.19)***	1.26 (0.96–1.66)	3.18 (1.55–6.53)**	3.61 (2.33–5.60)***	1.06 (0.41–2.74)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.419**

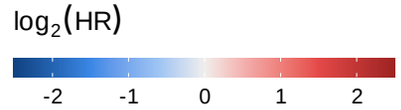


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR including patients flagged for deletion

n = 1797

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.14 (0.96–1.37)	1.09 (0.87–1.35)	1.69 (0.87–3.27)	1.15 (0.74–1.80)	1.01 (0.53–1.93)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.57 (1.21–2.02)***	1.65 (1.22–2.25)**	2.55 (1.02–6.35)*	1.43 (0.75–2.74)	1.01 (0.34–3.02)
EH-TG	1.79 (1.38–2.33)***	1.57 (1.11–2.24)*	4.51 (2.15–9.44)***	2.04 (1.14–3.65)*	1.10 (0.36–3.37)
EH-NL	1.25 (1.03–1.52)*	0.84 (0.64–1.10)	2.98 (1.59–5.60)***	1.94 (1.26–3.00)**	1.68 (0.88–3.21)
EH-TG+NL	1.68 (1.08–2.62)*	1.14 (0.59–2.19)	4.16 (1.14–15.16)*	2.36 (1.01–5.55)*	4.16 (1.16–14.94)*
Sys-TG	1.99 (1.40–2.82)***	1.55 (0.95–2.53)	1.92 (0.54–6.89)	4.13 (2.23–7.66)***	NE (1 event)
Sys-NL	1.68 (1.27–2.22)***	1.03 (0.68–1.55)	5.89 (2.74–12.66)***	3.31 (1.92–5.70)***	NE (2 events)
Sys-TG+NL	1.73 (1.41–2.13)***	1.24 (0.94–1.63)	3.20 (1.56–6.56)**	3.55 (2.29–5.51)***	1.06 (0.41–2.76)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.392**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR with the earlier pattern rule

STG + TTU and EHG + TTU classified as Sys-TG instead of Sys-TG+NL (22 patients reclassified)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.17 (0.98–1.40)	1.11 (0.89–1.38)	1.68 (0.87–3.26)	1.16 (0.74–1.82)	1.02 (0.53–1.95)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.58 (1.23–2.05)***	1.66 (1.22–2.25)**	2.53 (1.01–6.30)*	1.40 (0.73–2.69)	1.00 (0.33–3.00)
EH-TG	1.84 (1.41–2.39)***	1.58 (1.11–2.25)*	4.47 (2.14–9.36)***	2.01 (1.12–3.61)*	1.08 (0.35–3.32)
EH-NL	1.26 (1.03–1.53)*	0.84 (0.64–1.10)	3.01 (1.60–5.66)***	1.95 (1.27–3.01)**	1.68 (0.88–3.20)
EH-TG+NL	1.72 (1.10–2.68)*	1.14 (0.59–2.20)	4.09 (1.12–14.92)*	2.33 (0.99–5.48)	4.08 (1.14–14.67)*
Sys-TG	2.18 (1.62–2.95)***	1.76 (1.18–2.62)**	2.23 (0.71–6.94)	4.19 (2.38–7.38)***	NE (1 event)
Sys-NL	1.70 (1.29–2.26)***	1.03 (0.68–1.56)	5.86 (2.73–12.58)***	3.31 (1.92–5.69)***	NE (2 events)
Sys-TG+NL	1.71 (1.38–2.13)***	1.19 (0.89–1.59)	3.17 (1.53–6.57)**	3.58 (2.29–5.59)***	1.14 (0.44–2.95)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.397**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.